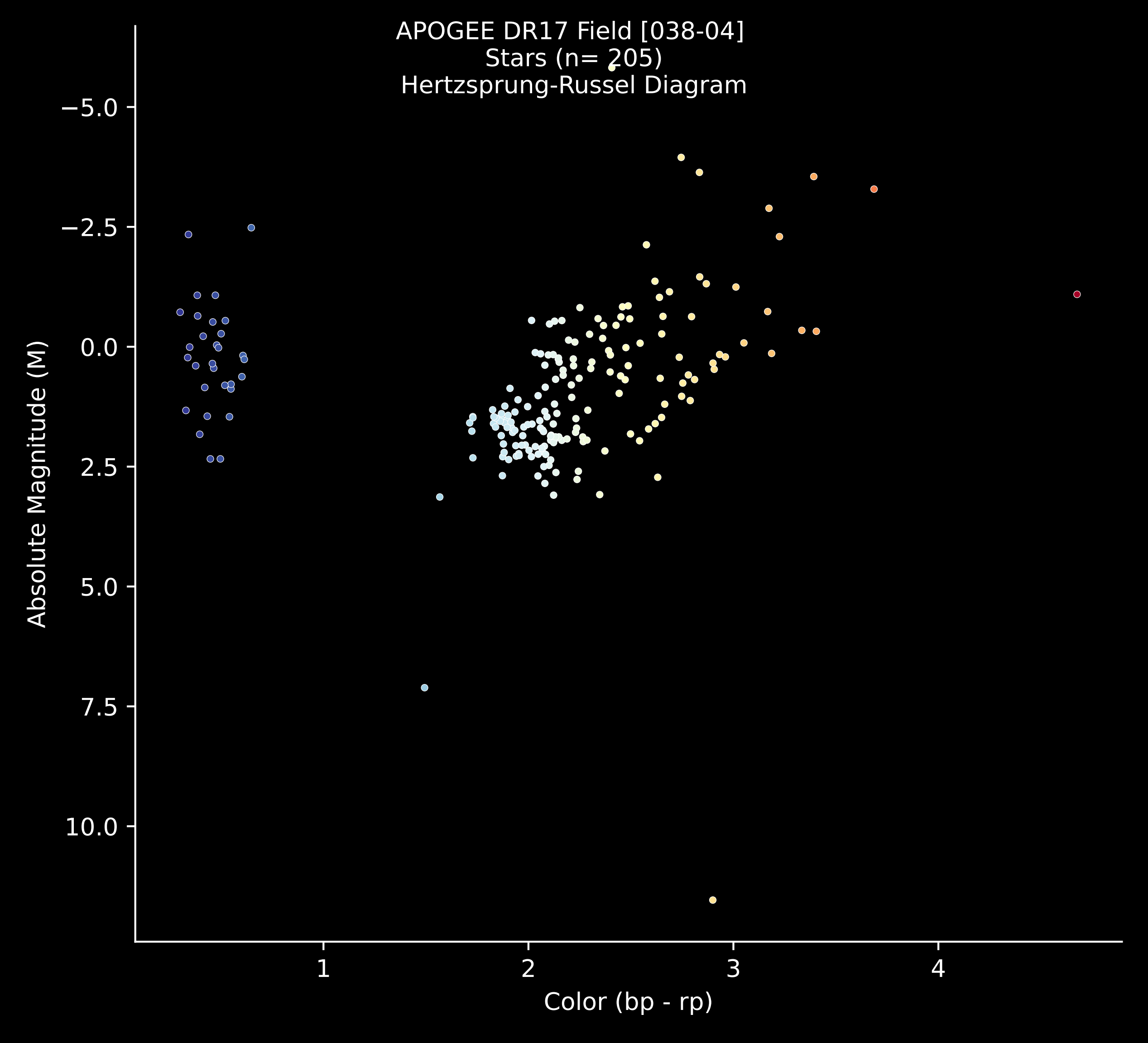
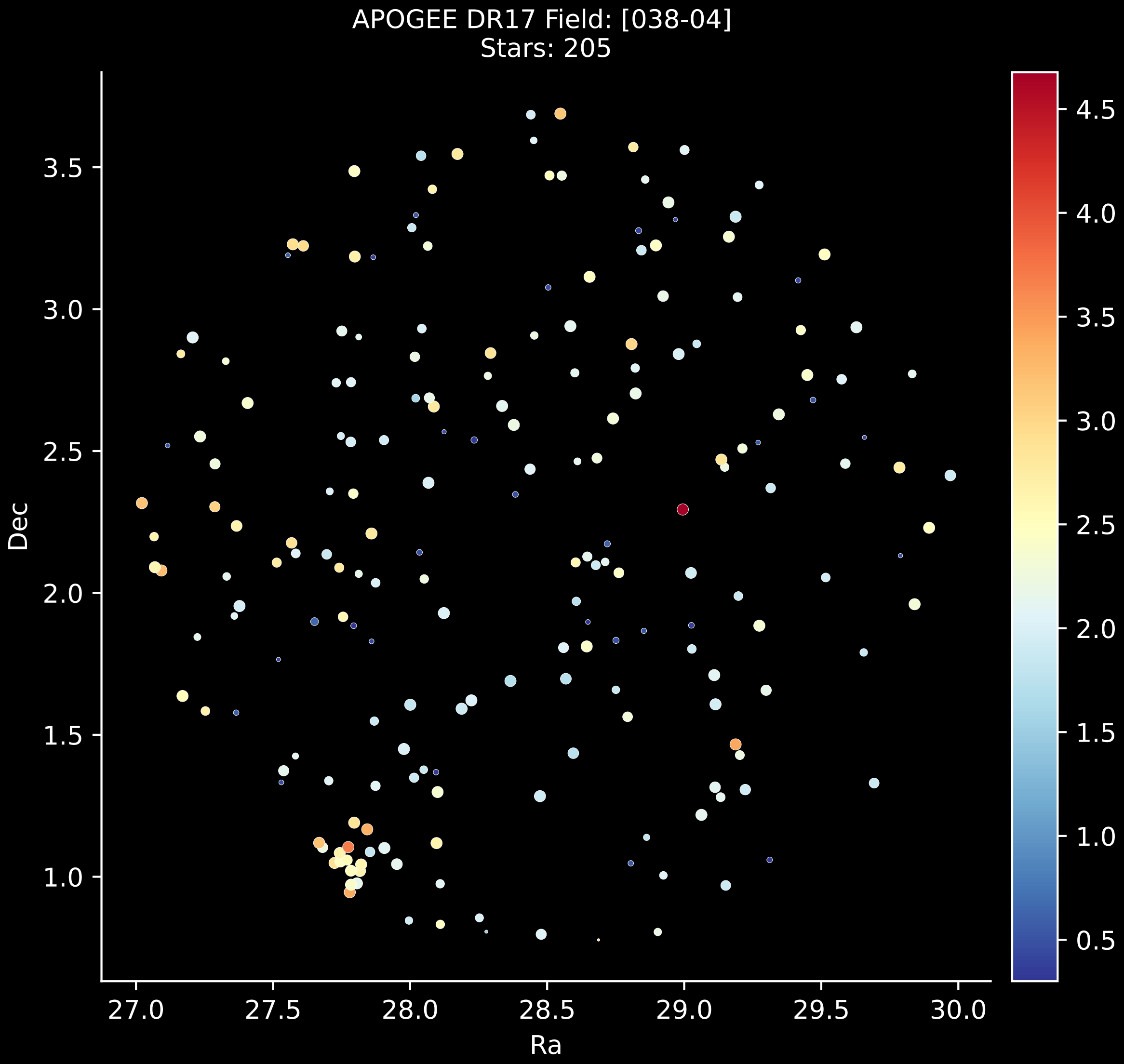
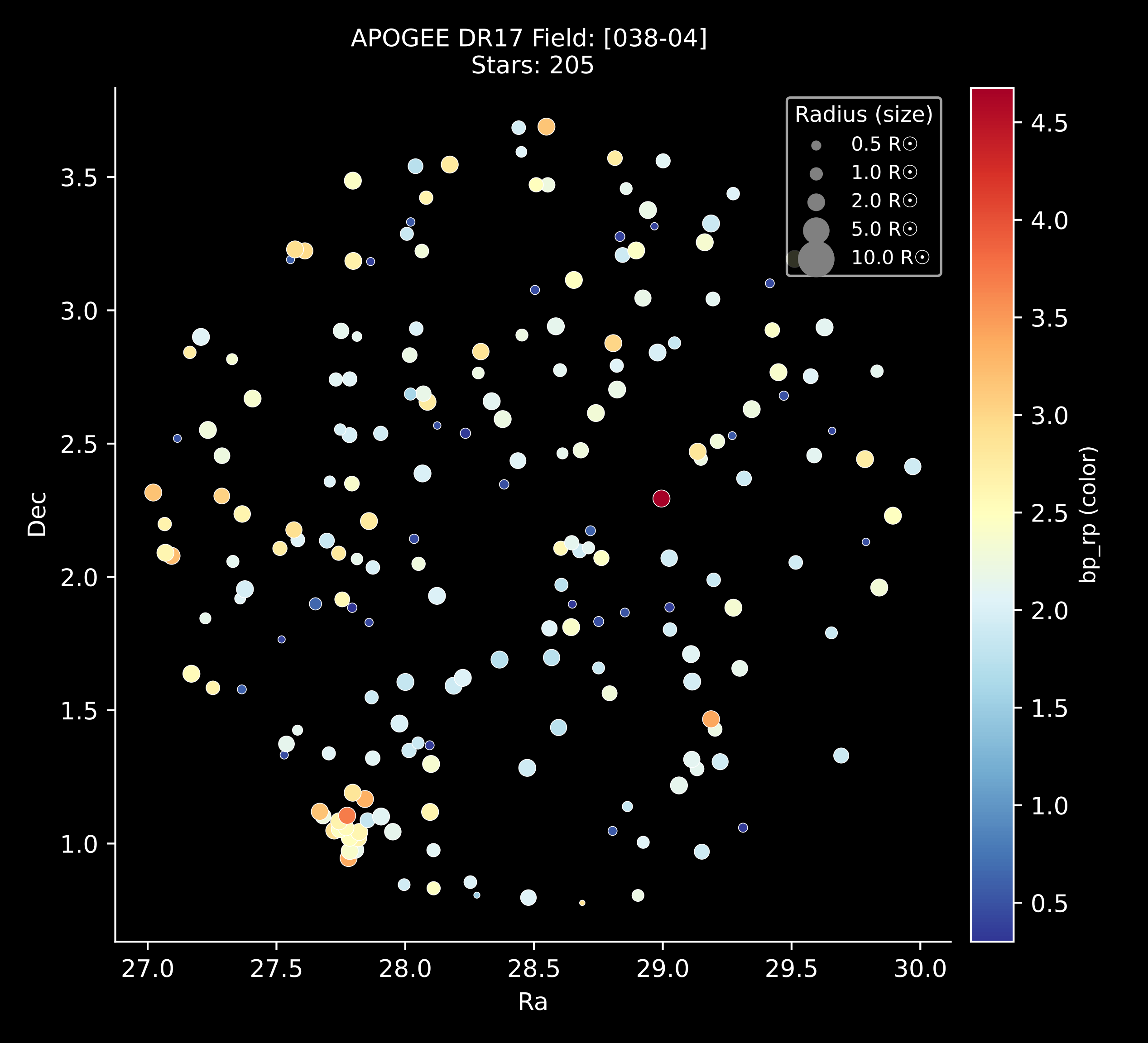
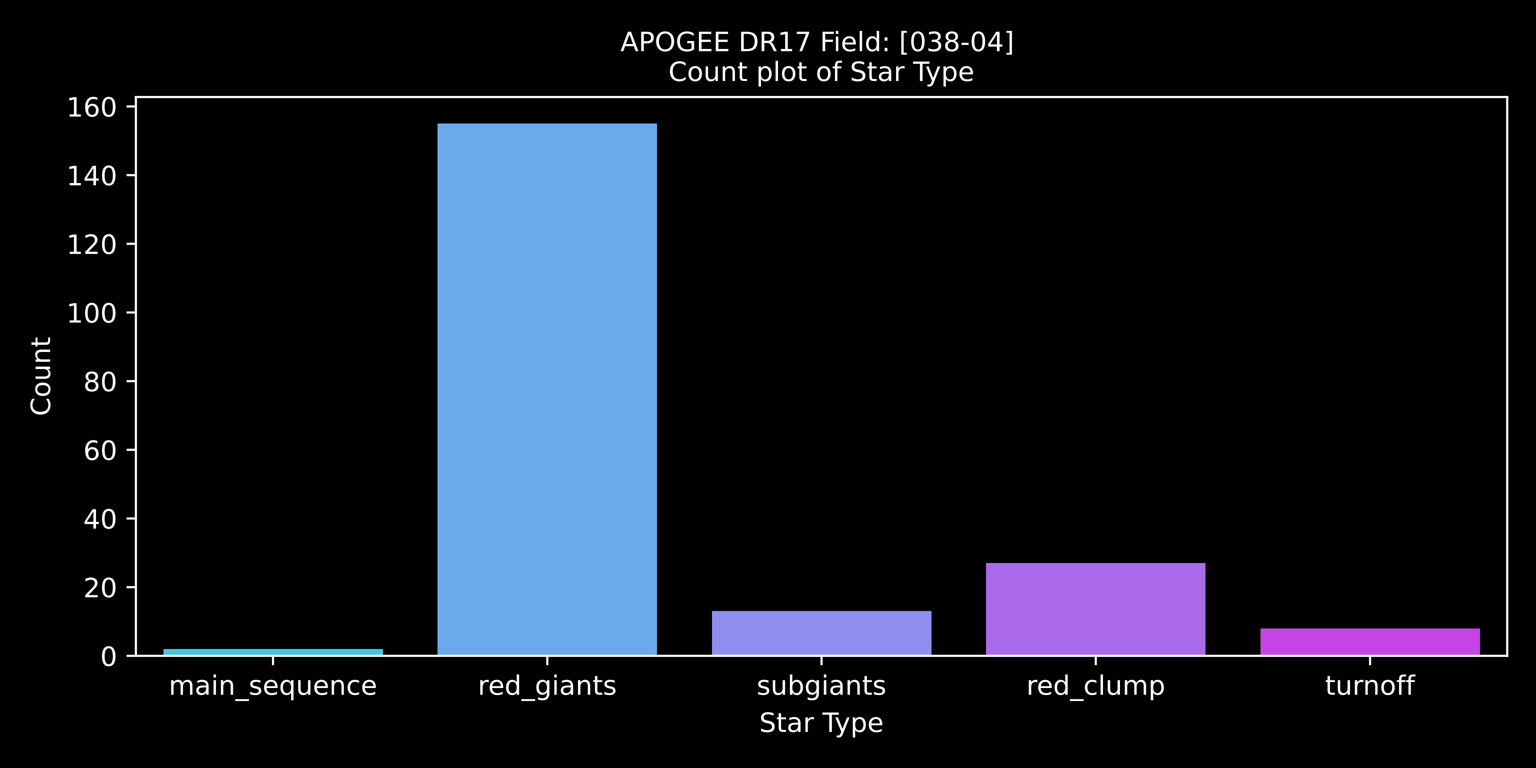


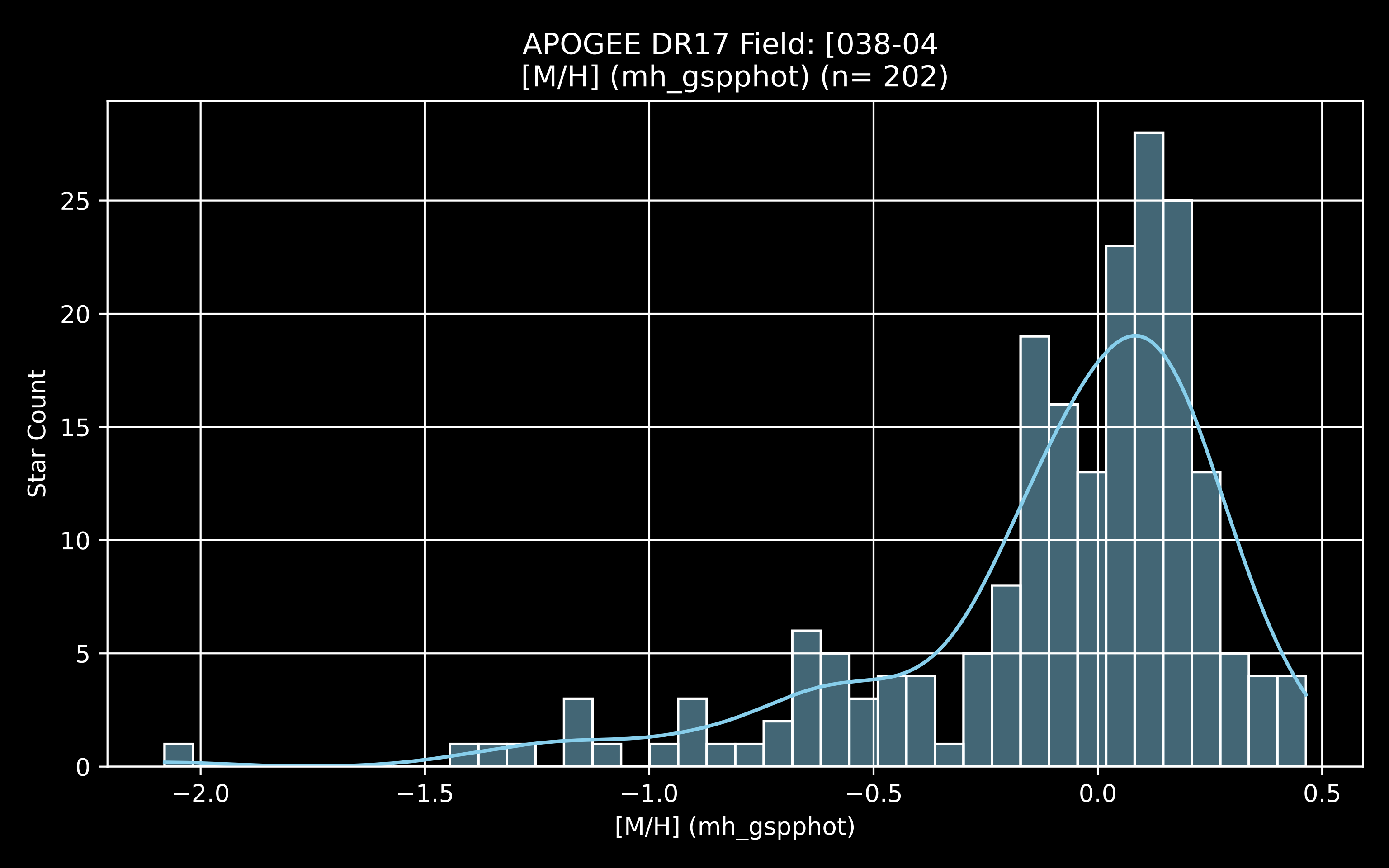
APOGEE DR17 Field [038-04]
Stars (n= 205)
Hertzsprung-Russel Diagram
Absolute Magnitude (M)
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
Color (bp - rp)
1
2
3
4

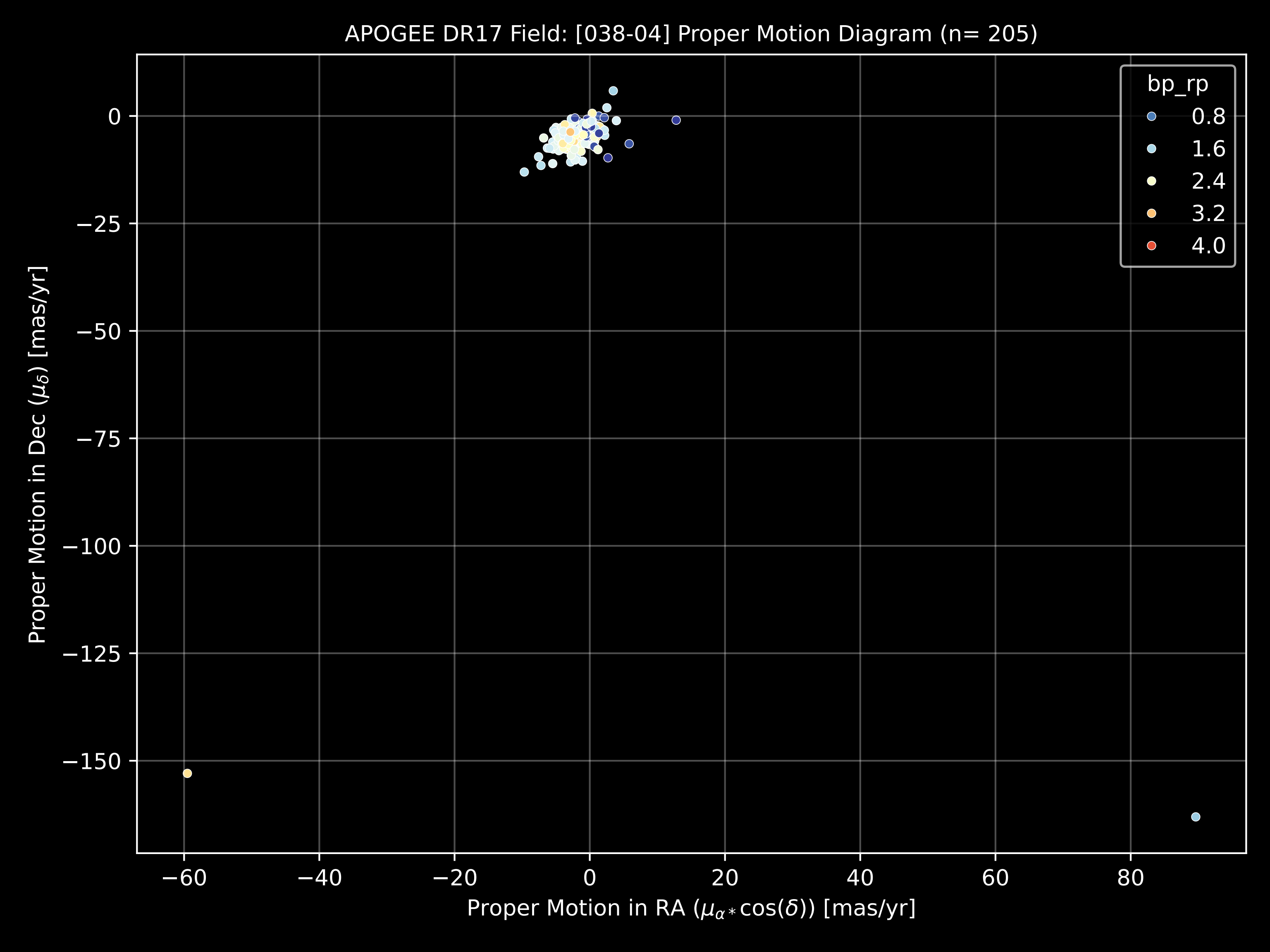

APOGEE DR17 Field: [038-04]
Stars: 205
Dec
Ra
bp_rp
3.5
3.0
2.5
2.0
1.5
1.0
27.0
27.5
28.0
28.5
29.0
29.5
30.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5



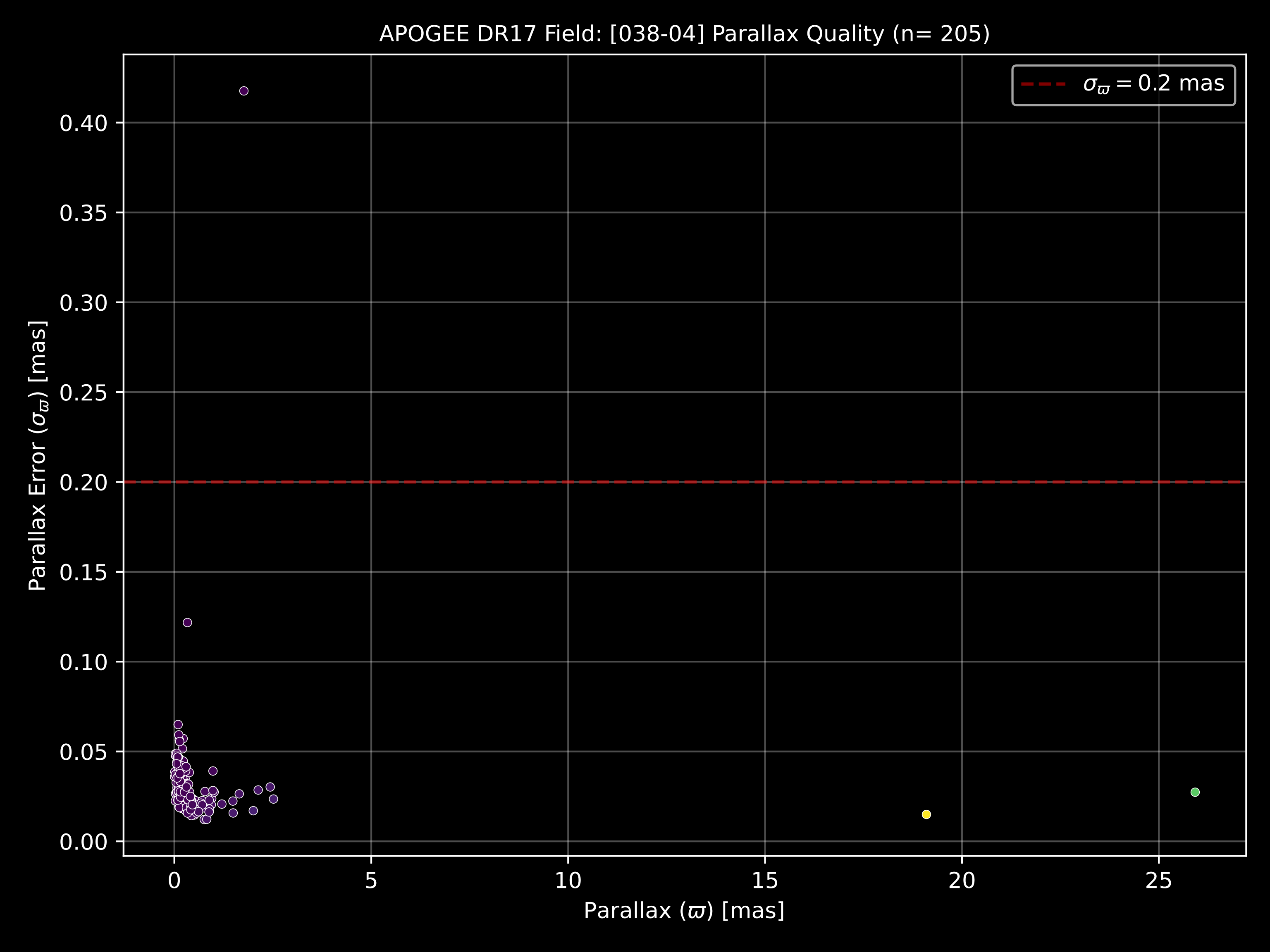
APOGEE DR17 Field: [038-04]
Stars: 205
Radius (size)
0.5 R☉
1.0 R☉
2.0 R☉
5.0 R☉
10.0 R☉
Dec
Ra
bp_rp (color)
3.5
3.0
2.5
2.0
1.5
1.0
27.0
27.5
28.0
28.5
29.0
29.5
30.0
4.5
4.0
0.5


APOGEE DR17 Field: [038-04]
Count plot of Star Type
Count
Star Type
0
20
40
60
80
100
120
140
160
main_sequence
red_giants
subgiants
red_clump
turnoff

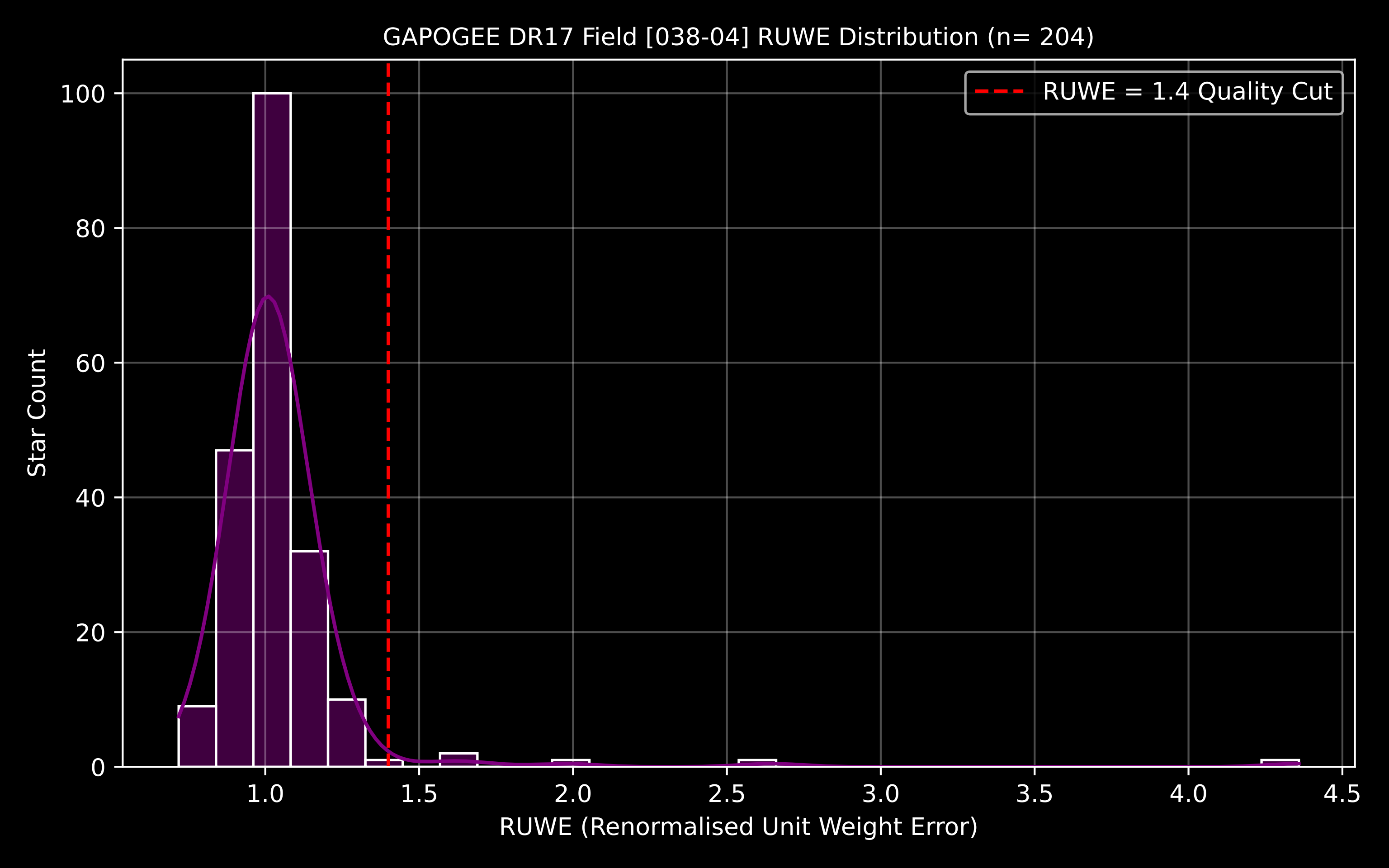

APOGEE DR17 Field: [038-04
[M/H] (mh_gspphot) (n= 202)
Star Count
[M/H] (mh_gspphot)
0
5
10
15
20
25
−2.0
−1.5
−1.0
−0.5
0.0
0.5

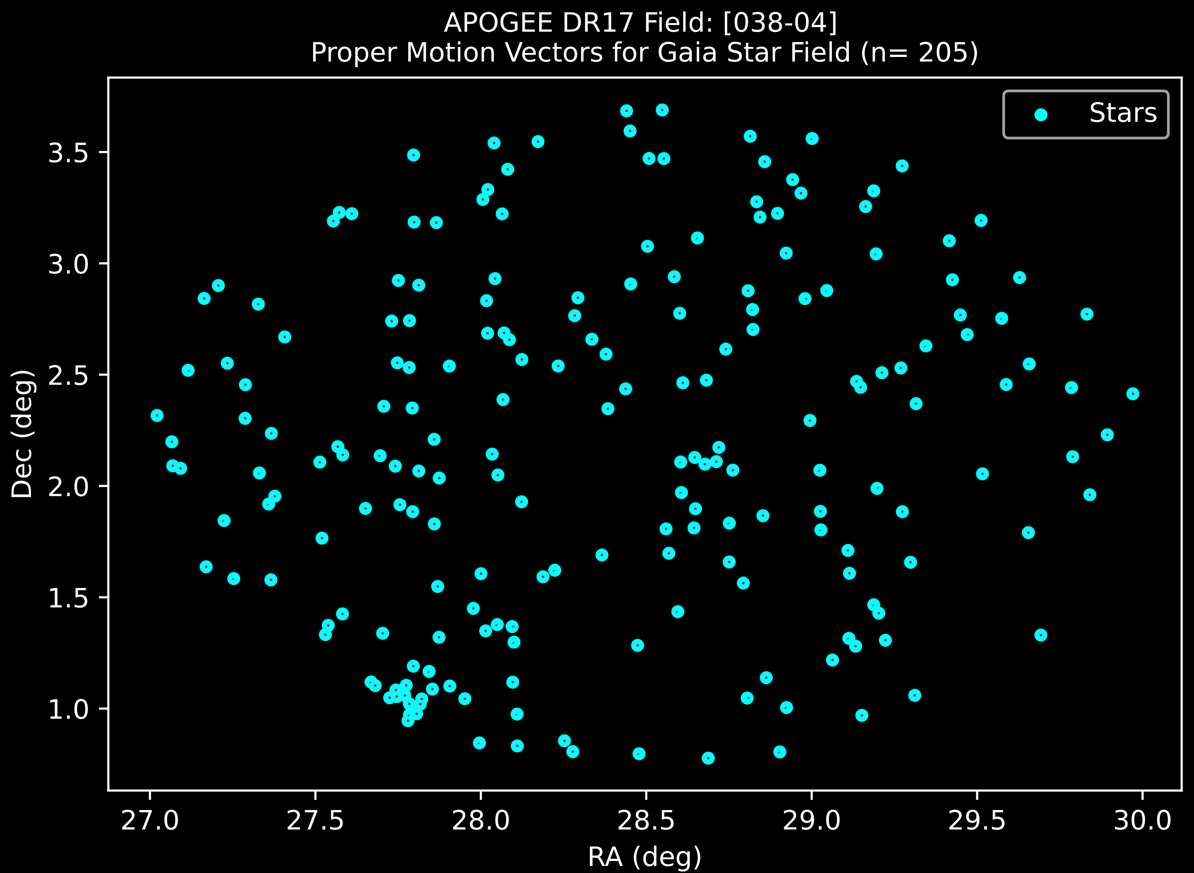


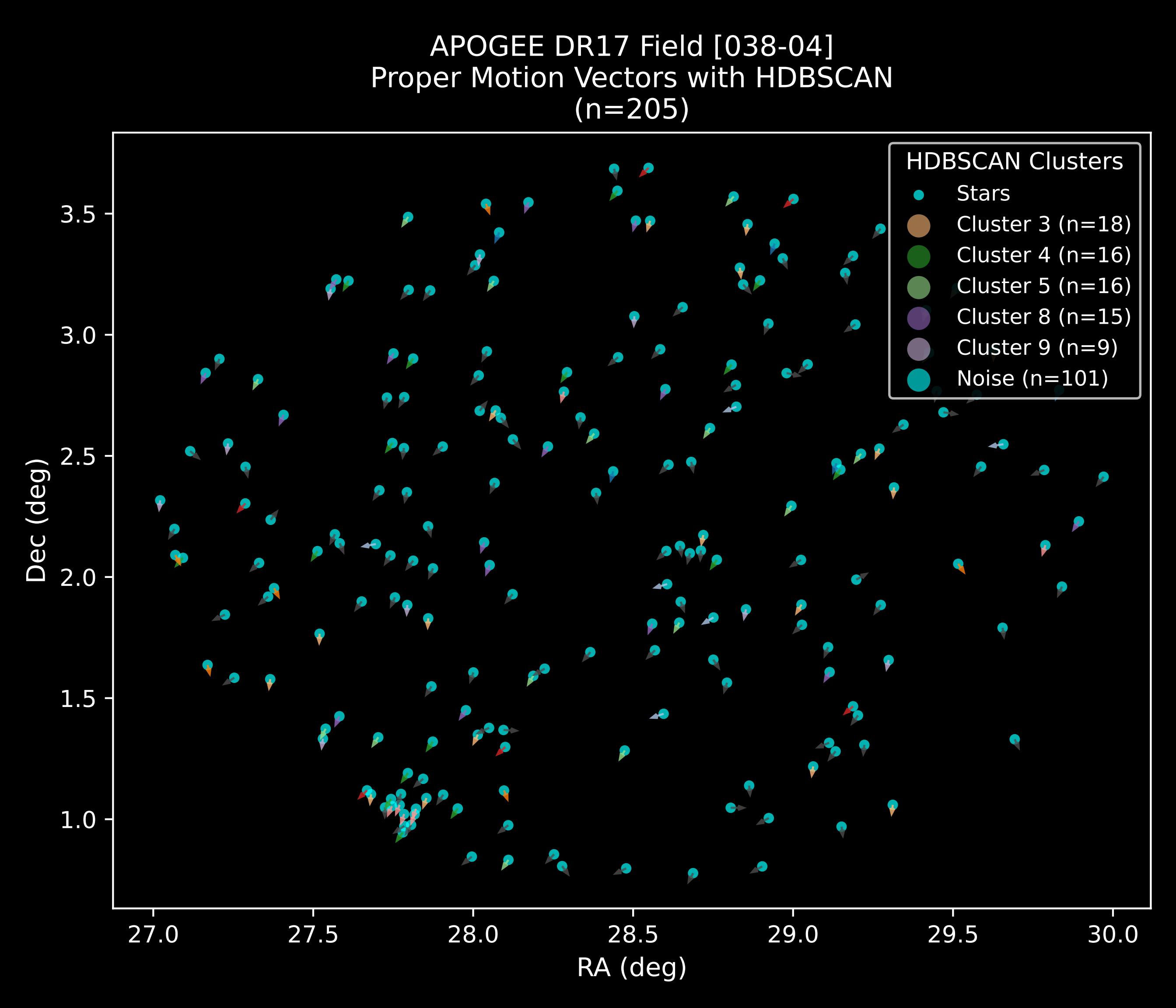
APOGEE DR17 Field: [038-04] Proper Motion Diagram (n= 205)
bp_rp
0.8
1.6
2.4
3.2
4.0
0
−25
−50
−75
−100
−125
−150
−60
−40
−20
0
20
40
60
80
Proper Motion in Dec (μ_δ) [mas/yr]
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]


APOGEE DR17 Field: [038-04] Parallax Quality (n= 205)
$\sigma_{\varpi} = 0.2$ mas
Parallax Error (σ_{ϖ}) [mas]
Parallax (ϖ) [mas]
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0
5
10
15
20
25

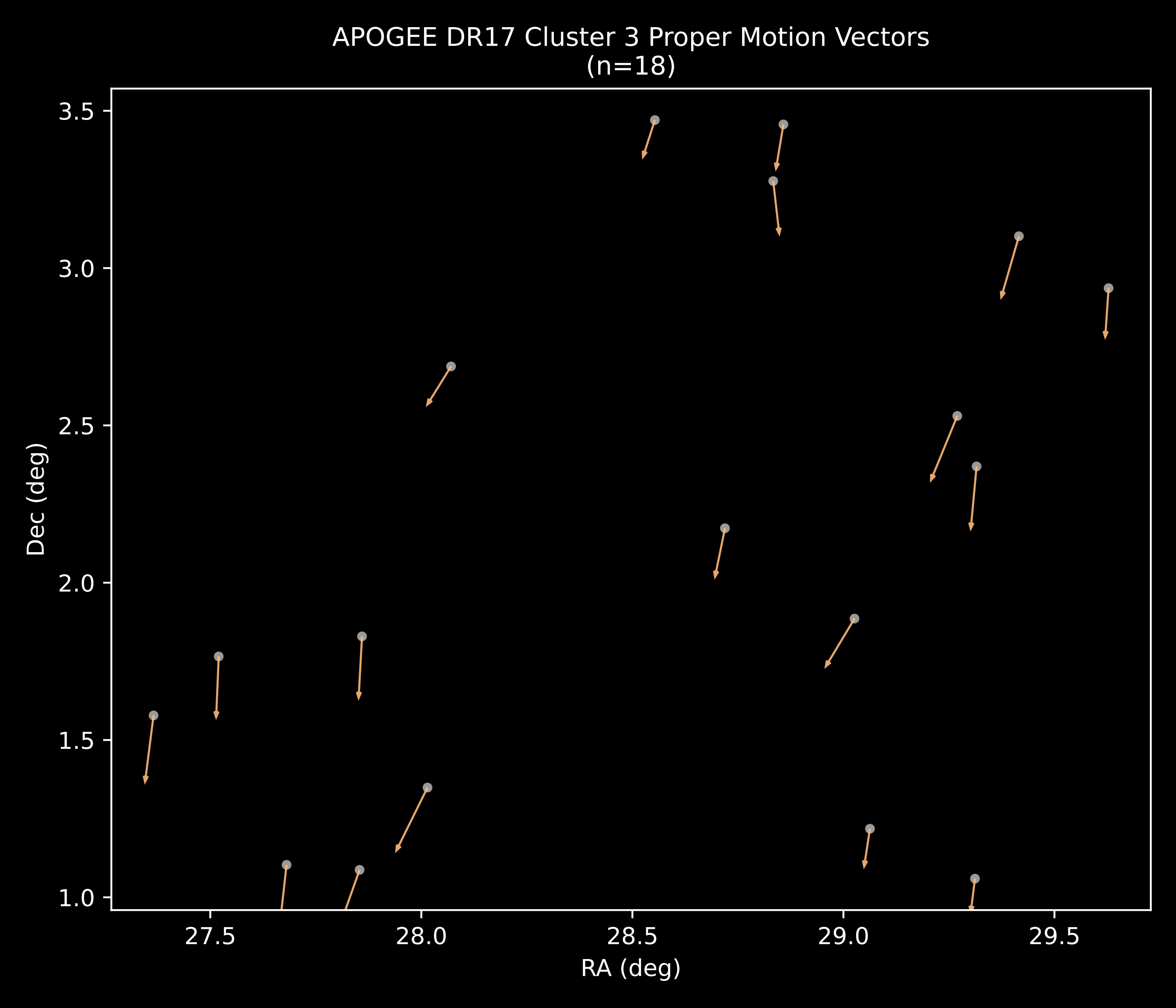
APOGEE DR17 Field: [038-04] Stellar Population Parameters (n= 205)
Metallicity ([M/H]) (gspphot)
Effective Temperature (T_{eff}) [K]
0.5
0.0
−0.5
−1.0
−1.5
−2.0
22500
20000
17500
15000
12500
10000
7500
5000
2500
logg_gspphot
1
2
3
4
5

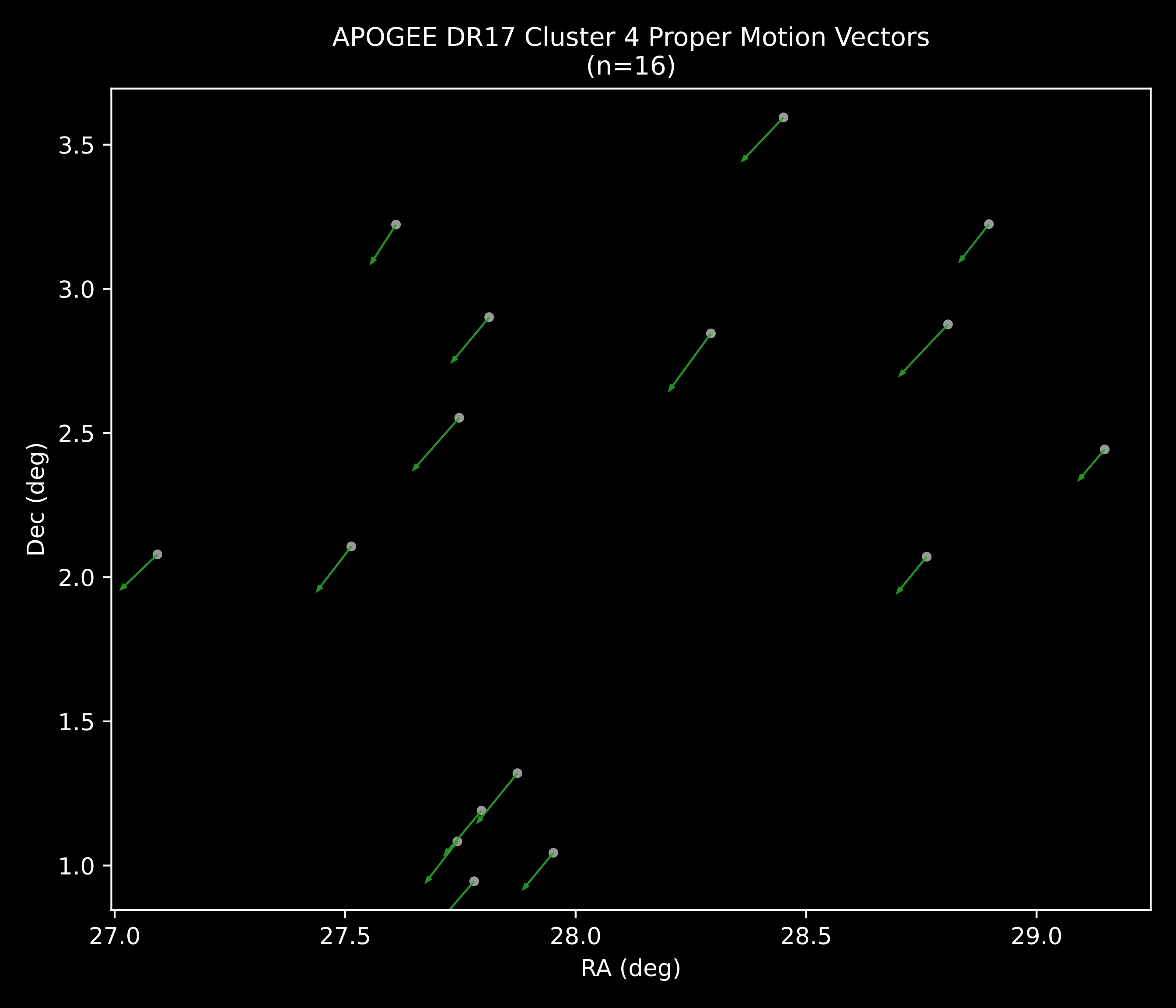

GAPOGEE DR17 Field [038-04] RUWE Distribution (n= 204)
RUWE = 1.4 Quality Cut
Star Count
RUWE (Renormalised Unit Weight Error)
0
20
40
60
80
100
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5


APOGEE DR17 Field: [038-04]
Proper Motion Vectors for Gaia Star Field (n= 205)
Stars
Dec (deg)
RA (deg)
3.5
3.0
2.5
2.0
1.5
1.0
27.0
27.5
28.0
28.5
29.0
29.5
30.0

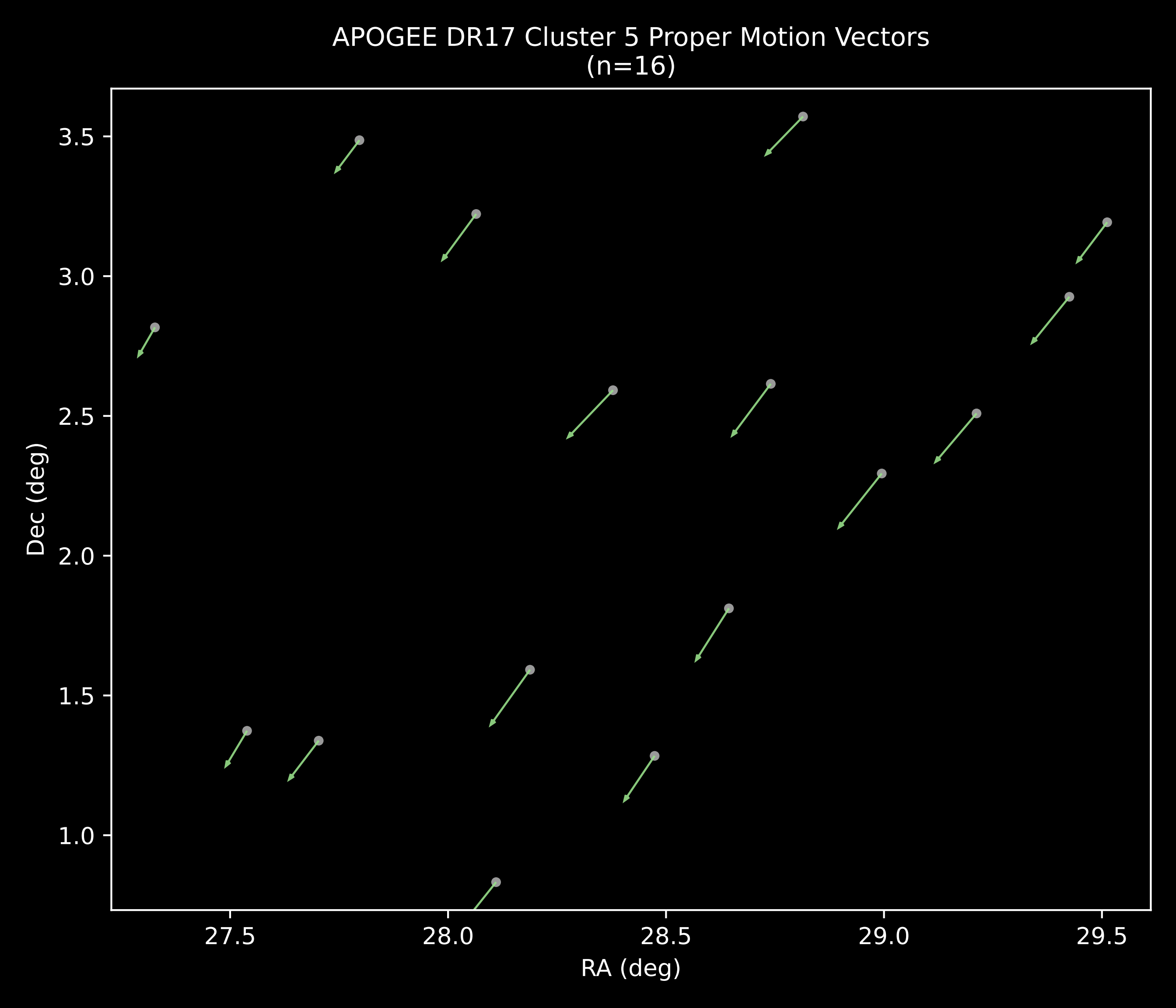


APOGEE DR17 Field [038-04]
Proper Motion Vectors with HDBSCAN
(n=205)
HDBSCAN Clusters
Stars
Cluster 3 (n=18)
Cluster 4 (n=16)
Cluster 5 (n=16)
Cluster 8 (n=15)
Cluster 9 (n=9)
Noise (n=101)
Dec (deg)
RA (deg)
3.5
3.0
2.5
2.0
1.5
1.0
27.0
27.5
28.0
28.5
29.0
29.5
30.0

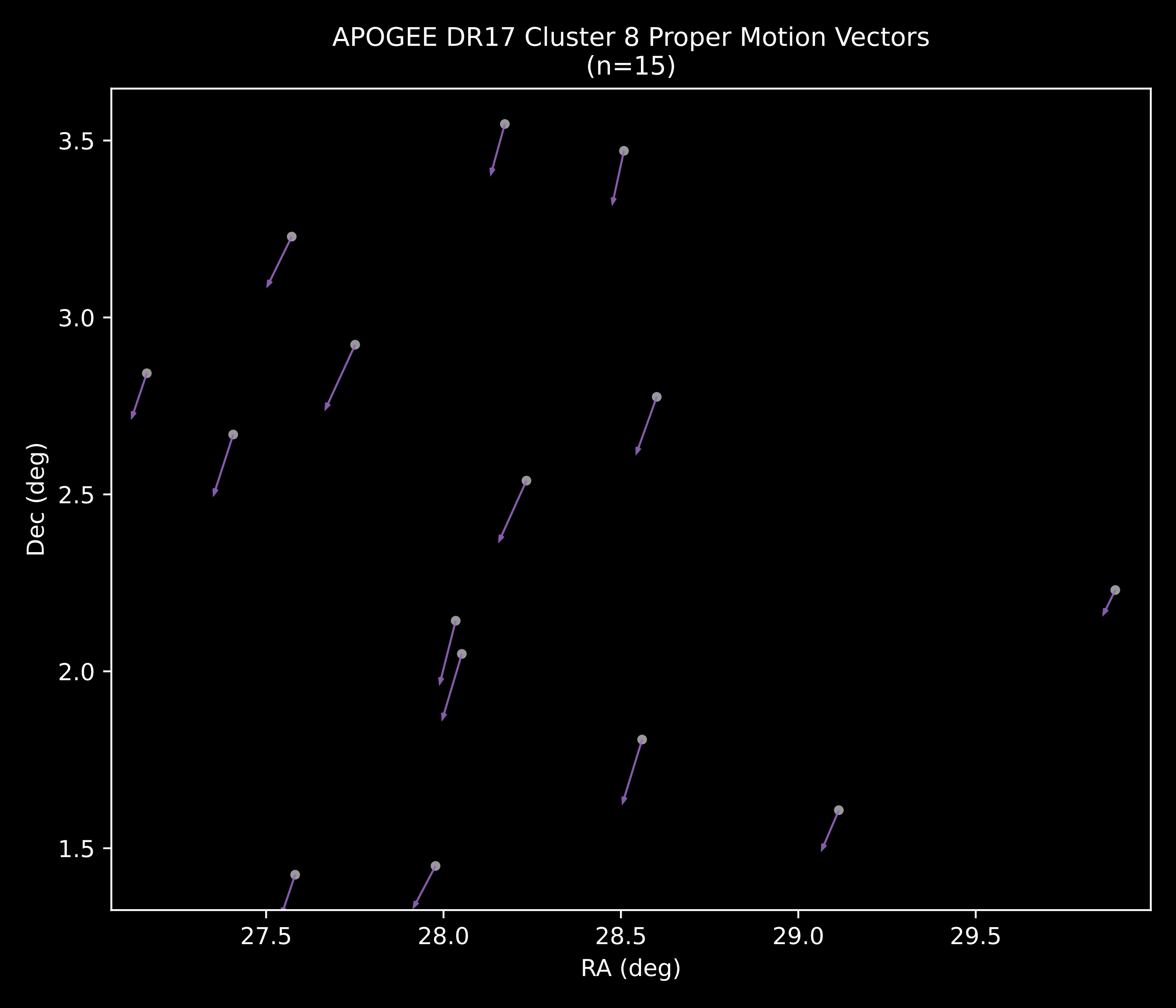

APOGEE DR17 Cluster 3 Proper Motion Vectors
(n=18)
3.5
3.0
2.5
2.0
1.5
1.0
Dec (deg)
27.5
28.0
28.5
29.0
29.5
RA (deg)

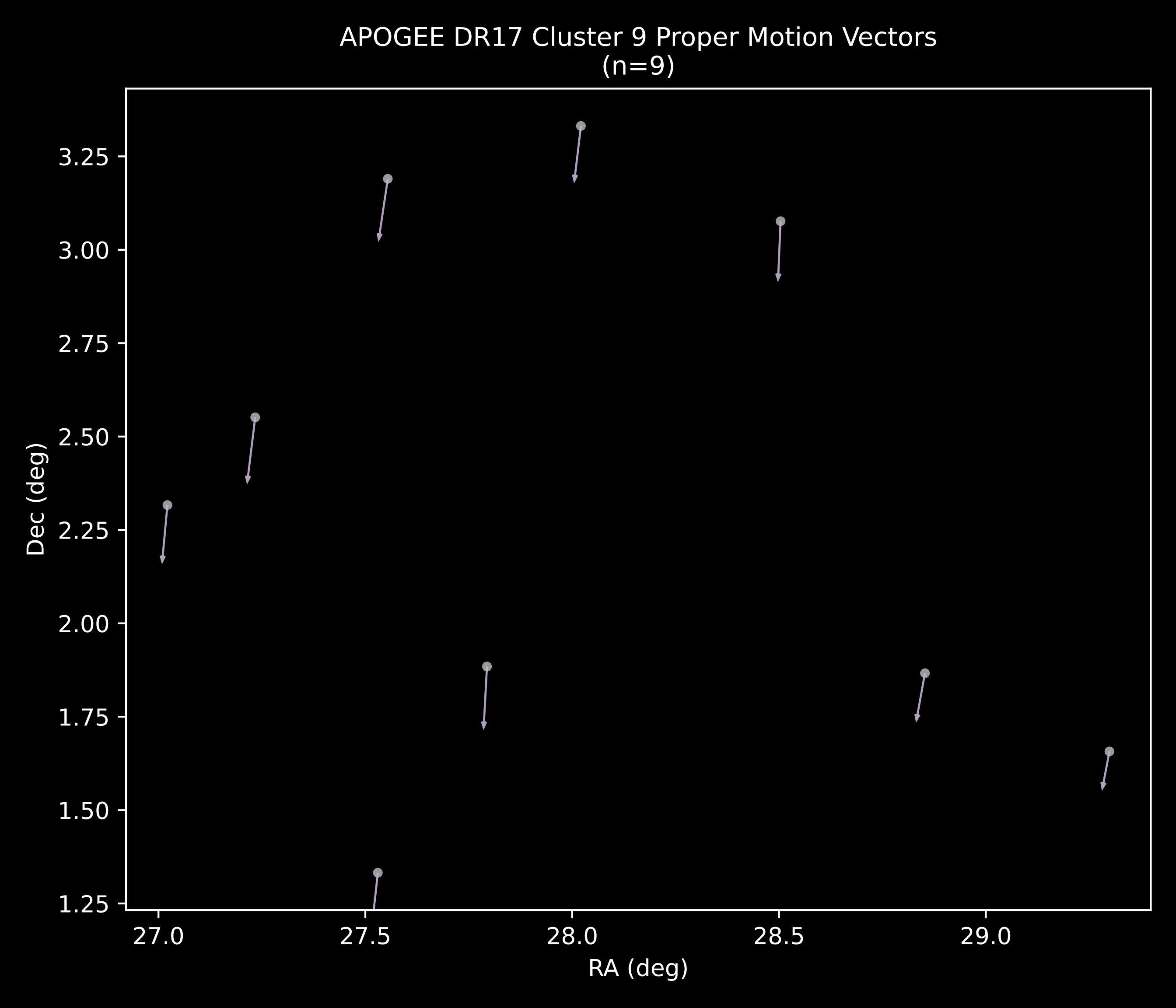


APOGEE DR17 Cluster 4 Proper Motion Vectors
(n=16)
3.5
3.0
2.5
2.0
1.5
1.0
Dec (deg)
27.0
27.5
28.0
28.5
29.0
RA (deg)

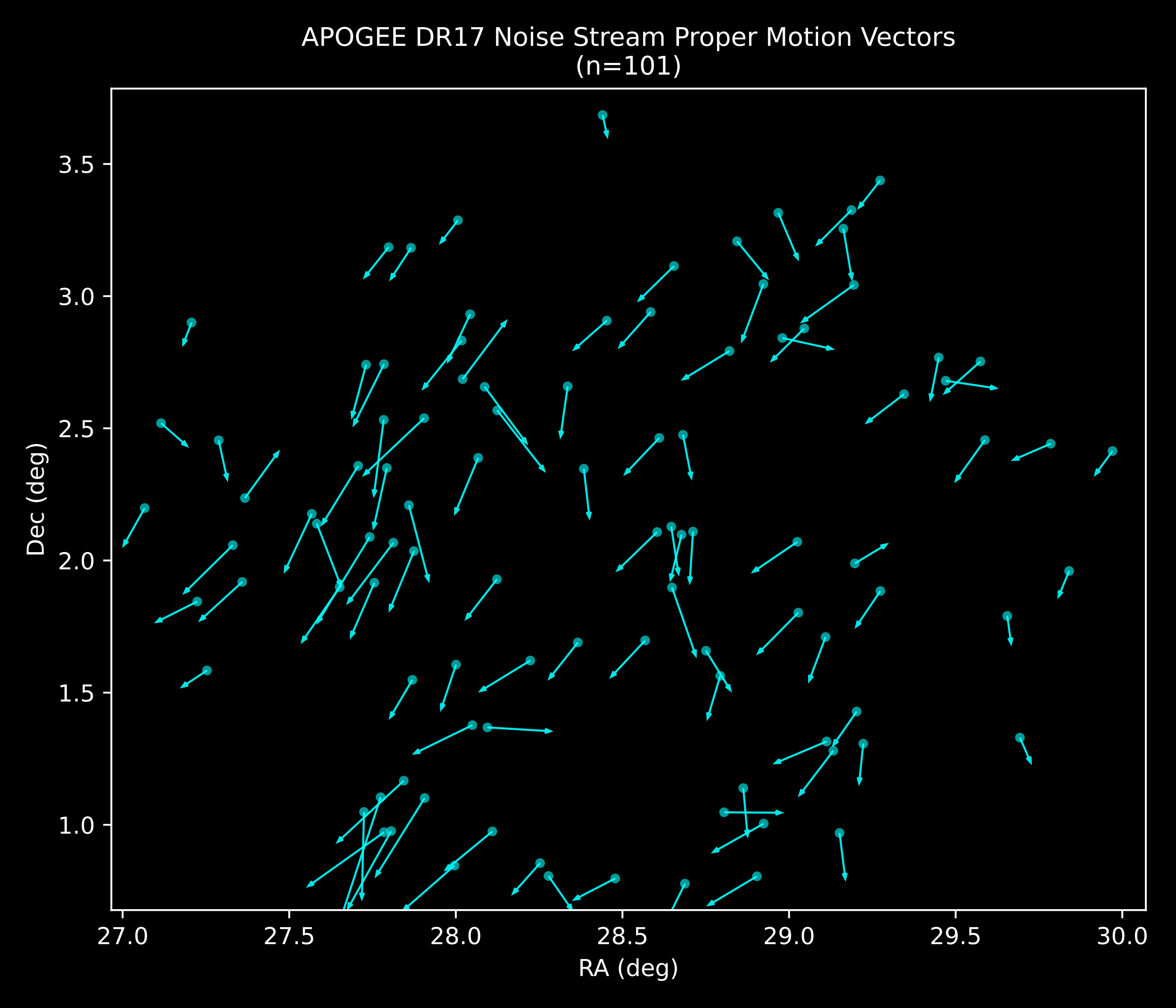


APOGEE DR17 Cluster 5 Proper Motion Vectors
(n=16)
Dec (deg)
RA (deg)
3.5
3.0
2.5
2.0
1.5
1.0
27.5
28.0
28.5
29.0
29.5


APOGEE DR17 Cluster 8 Proper Motion Vectors
(n=15)
3.5
3.0
2.5
2.0
1.5
Dec (deg)
27.5
28.0
28.5
29.0
29.5
RA (deg)



APOGEE DR17 Cluster 9 Proper Motion Vectors
(n=9)
Dec (deg)
RA (deg)
3.25
3.00
2.75
2.50
2.25
2.00
1.75
1.50
1.25
27.0
27.5
28.0
28.5
29.0


APOGEE DR17 Noise Stream Proper Motion Vectors
(n=101)
Dec (deg)
RA (deg)
3.5
3.0
2.5
2.0
1.5
1.0
27.0
27.5
28.0
28.5
29.0
29.5
30.0